

Supertails Data Analytics Assessment Report

1. Introduction

This assessment analyzes Supertails’ end-to-end e-commerce, communication, supply chain, and customer support ecosystem using five datasets: **orders, communication logs, supply chain, support tickets, and vet consultations**.

The objective is to evaluate operational efficiency, customer behavior, communication effectiveness, and support outcomes using **SQL, Python analytics, customer segmentation, and predictive modeling**.

The analysis covers:

- 1. Data cleaning and validation
- 2. Exploratory Data Analysis (EDA)
- 3. RFM-based customer segmentation
- 4. Predictive modeling for support ticket likelihood

2. Data Preparation & Validation

2.1 Dataset Overview

Dataset	Rows
Orders	~10,000
Communication Logs	~40,000
Supply Chain	~10,000
Support Tickets	~5,000
Vet Calls	~5,000

2.2 Data Cleaning & Validation Metrics

- **Missing values:** 0 missing values in primary identifiers across all datasets
- **Date consistency checks:**
 - 100% of actual_delivery_date ≥ order_date
 - 100% of resolved_at ≥ created_at
- **Supply chain event order:**
 - ~94% followed correct scan order
 - ~6% exhibited scan irregularities (flagged, not dropped)
- **Foreign key integrity:**

- 100% of order_id references across all datasets matched the orders table
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3. Exploratory Data Analysis (EDA)

3.1 Order Value Distribution

- Order values ranged broadly with **no heavy skew**
- Mean order value was consistent across bins
- No dominance of low-cost or luxury purchases

Insight:

Customers purchase across a wide price range, indicating a balanced mix of budget and premium demand.

3.2 Category-wise Revenue

- Revenue contribution was relatively uniform across categories
- **Grooming** emerged as the highest revenue contributor
- **Food** contributed the least
- Accessories and veterinary pharmaceuticals showed similar revenue levels

Insight:

Revenue is diversified across product categories, reducing dependency on a single category.

3.3 City-level Performance

- All cities generated **₹2.5M+ revenue**
- No single city disproportionately dominated GMV
- Revenue distribution across cities was balanced

Insight:

Demand is geographically diversified, indicating stable regional performance.

3.4 Payment Mode Breakdown

- Digital payments (UPI, Cards, Wallets) contributed **>75% of total revenue**
- Cash on Delivery contributed comparatively less

Insight:

Strong digital payment adoption reduces operational friction and cash handling risk.

3.5 Delivery Turnaround Time (TAT)

- Delivery TAT ranged from **1 to 7 days**
- Distribution was nearly uniform
- No significant long-tail delays observed

Insight:

Delivery operations are stable with minimal extreme delays.

3.6 Communication Channel Effectiveness

Metrics were calculated as **action ÷ delivered messages**.

Channel	Read Rate	Click Rate
WhatsApp	~0.37	~0.21
Email	~0.35	~0.19
SMS	~0.34	~0.18
Call	~0.38	~0.22

Insights:

- Engagement rates were tightly clustered across channels
 - Calls and WhatsApp showed marginally higher engagement
 - No channel significantly outperformed others
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3.7 Support Ticket Trend by Week

- Weekly ticket volume ranged between **~75 and ~120 tickets**
- No sustained upward or downward trend
- A few isolated peaks and dips were observed

Insight:

Customer support demand remained stable throughout the year, indicating consistent operations.

4. RFM Analysis & Customer Segmentation

4.1 RFM Metrics

- **Recency:** Days since last purchase
- **Frequency:** Total orders per customer
- **Monetary:** Total spend per customer

Quantile-based scoring (with duplicate handling) was used to generate R, F, and M scores.

4.2 Segment Distribution

Segment Customers

Bronze 3,237

New 1,741

Silver 709

Gold 315

Insights:

- ~65% of customers fall into Bronze or New segments
- Gold customers represent ~6% of the customer base
- Distribution aligns with typical e-commerce behavior

4.3 Revenue Contribution by Segment

- **Bronze:** Highest total revenue due to large population
- **Silver & New:** Moderate revenue contribution
- **Gold:** Lower total revenue but highest **average revenue per customer**

Insight:

Revenue is volume-driven at the lower end, while high-value customers are few but strategically important.

5. Predictive Analytics – Support Ticket Prediction

5.1 Model Overview

- **Model:** Logistic Regression
- **Target:** Customer raises a support ticket (Yes/No)
- **Features:**
 - Average order value
 - Repeat customer status
 - City
 - Shipment mode
 - Communication click & reply rates

5.2 Model Performance Metrics

Metric	Value
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Accuracy	~0.68
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Recall	~0.61
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F1 Score	~0.64
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ROC-AUC	~0.71
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(Exact values may vary slightly due to random split)

5.3 Business Interpretation

- Communication engagement and repeat behavior influence support ticket likelihood
 - Model shows reasonable discrimination capability
 - Enables **proactive support identification** and workload planning
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6. Key Business Takeaways

- Revenue and demand are geographically and category-wise diversified
 - Communication channels perform comparably, enabling flexible outreach strategies
 - Delivery operations are stable with predictable timelines
 - Majority of customers are low-engagement, creating retention opportunities
 - Gold customers are few but high-impact
 - Predictive modeling enables proactive customer support optimization
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7. Conclusion

This analysis provides a comprehensive, metric-driven understanding of Supertails' operations and customer behavior. By combining SQL, Python analytics, RFM segmentation, and predictive modeling, the assessment demonstrates a scalable, data-driven approach to improving revenue growth, customer retention, and operational efficiency.